Sub-Saharan Africa Agricultural & Food Security Risk Analysis (2000–2024)

Executive Summary

This project delivers a comprehensive longitudinal analysis of Sub-Saharan African agricultural metrics (Area harvested, Yield, and Production volume) using FAOSTAT data from 2000 to 2024. By standardizing units, cleaning data anomalies, and constructing multi-year moving averages alongside a Food Security Risk Indicator, this study identifies critical vulnerabilities in regional staple production.

Stage 1: Data Cleaning & Preparation

● **Data Sources:** FAOSTAT crops and livestock products dataset covering multiple Sub-Saharan African regions.
● **Standardization:** Converted all yield metrics into uniform units (t/ha) and standardized country and crop category names.
● **Derived Metrics:** Computed year-over-year (YoY) yield growth rates, 3-year moving averages to smooth out single-year weather anomalies, and helper columns for structural trend evaluation.

Stage 2: Regional Yield Analysis & Risk Rationale

Pivot tables were constructed to evaluate long-term yield trajectories across regions and crops:
● **Critical Risk — Western Africa (Fresh Cassava):** While experiencing a moderate 24-year yield contraction (-20.05%), Cassava serves as the primary caloric staple across Western Africa with a massive baseline production volume exceeding 79 million tonnes annually. Declining land yields at this scale pose severe risks to regional food supply stability.
● **High Risk — Southern Africa (Sorghum & Millet):** Southern Africa suffered severe, sustained yield collapses. Sorghum declined by -47.11% (from 2.34 t/ha to 1.24 t/ha), and Millet experienced a massive -79.79% drop, highlighting acute vulnerability to climate shocks, drought, and soil degradation.● **Low Risk / Stable Growth — Eastern & Middle Africa (Maize, Rice):** Demonstrated strong resilience, with regional maize and rice yields achieving positive long-term

expansion (e.g., Southern Africa maize growing by +70.24%).

Stage 3: Visualization & Dashboard Reporting

● **Dashboard Tab:** Built a centralized Google Sheets dashboard incorporating a 2000–2024 trend line chart mapping all regional crop trajectories.

● **Color-Coded Risk Table:** Summarized net growth percentages with conditional formatting (red for negative growth, green for positive growth) to highlight high-risk sectors instantly.

● **Expected Deliverables:**

1. Google Sheets Workbook (Raw data, cleaning log, pivot tables, and dashboard tab).
2. 1-Page Ranked Risk Summary with Supporting Rationale.
3. GitHub Repository integration with completed documentation and README.